PROGRAM: SpringBoard Internship Program / Infosys Internship

PROJECT: Personalized TED Talks Recommendation System Using Machine Learning

DELIVERABLE: Milestone 1 — Module 1 (Data Collection) & Module 2 (EDA & Preprocessing)

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GITHUB REPOSITORY: https://github.com/shaheed3515/ted-talks-recommendation-system

Module 1: Data Collection & Quality Audit Report

1. Module Objectives & Syllabus Requirements

As defined in the project syllabus (`ted.pdf`), Milestone 1 requires:

- **Data Collection:** Collect a high-quality dataset of TED Talks including metadata (titles, tags, speaker names, transcripts, and user views/ratings).
- **Diversity & Volume:** Ensure broad topic coverage across disciplines (technology, science, global issues, humanities) with sufficient volume for training content-based and collaborative recommendation engines.
- **Exploratory Data Analysis:** Rigorous examination of talk length distributions, topic tags, vocabulary sizes, and data quality.

2. Dataset Summary & Scale

2,550	13.8 min	7.5	2,044
TOTAL UNIQUE TALKS	AVERAGE TALK DURATION	AVG TOPICS / VIDEO	AVG WORDS / TALK

2.1 Collected Source Files

Source File	File Size	Record Count	Columns	Primary Content Captured
ted_main.csv	7.3 MB	2,550 talks	17 columns	Title, speaker, tags, views, duration, occupation, ratings, URL
transcripts.csv	28.0 MB	2,467 talks	2 columns	Complete verbatim spoken speech text, URL
Cleaned Master	69.0 MB	2,550 unique talks	26 columns	Merged metadata + transcripts + cleaned text + combined_content

3. Key Exploratory Analysis: Topics & Distributions

3.1 Topic Diversity (Topics per Video)

Each TED Talk is tagged with multiple conceptual labels (e.g. *technology, creativity, design, science*). Our analysis revealed:

- **Average topics per video:** 7.5 tags per talk (Median: 6 tags).
- **Topic range:** From 1 tag up to 32 tags per video.
- **Top 5 Categories:** Technology (727 talks), Science (567 talks), Global Issues (501 talks), Culture (486 talks), and Design (418 talks).

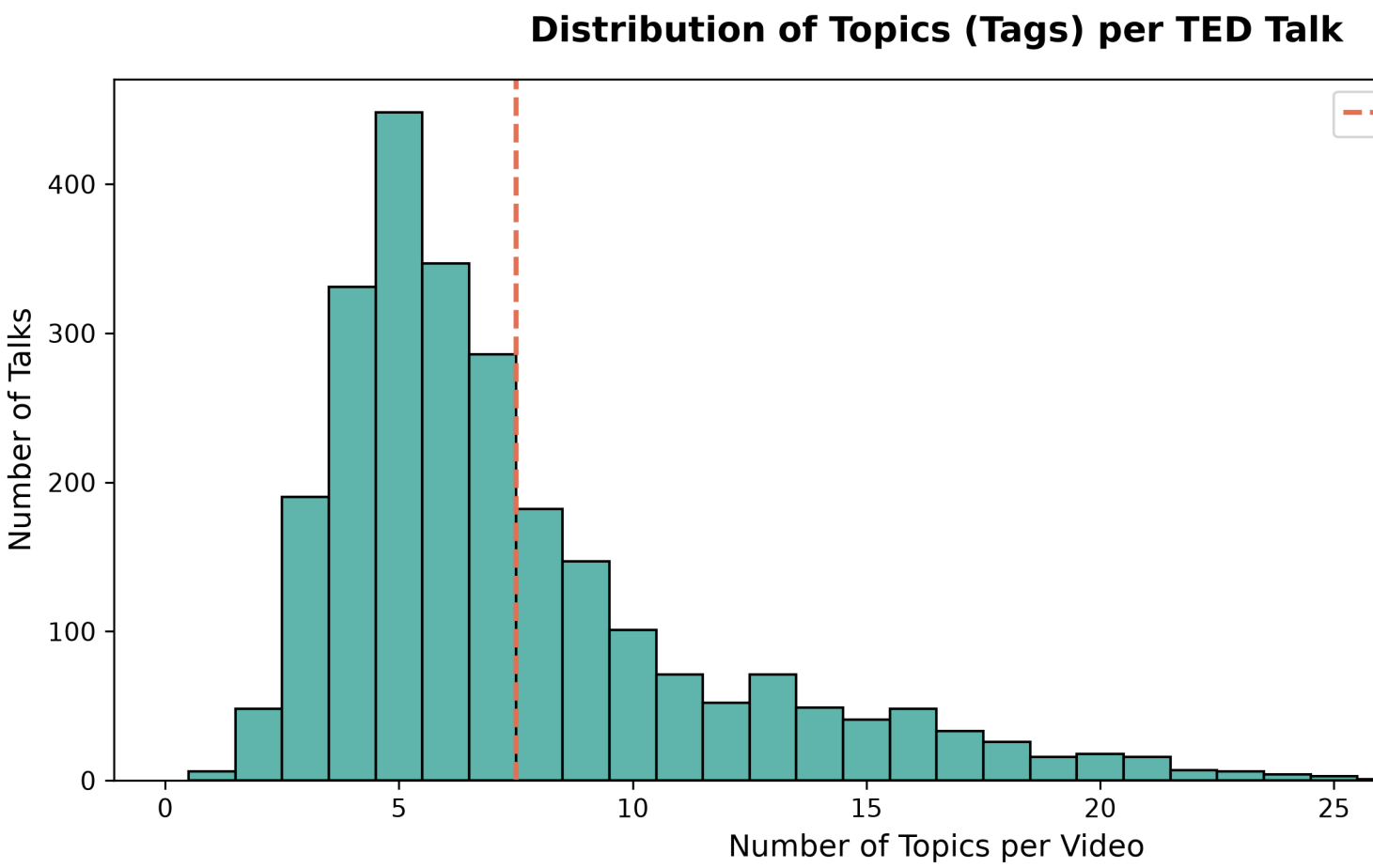


Figure 1: Distribution of Topic Tags per TED Talk (Mean: 7.5 topics/video).

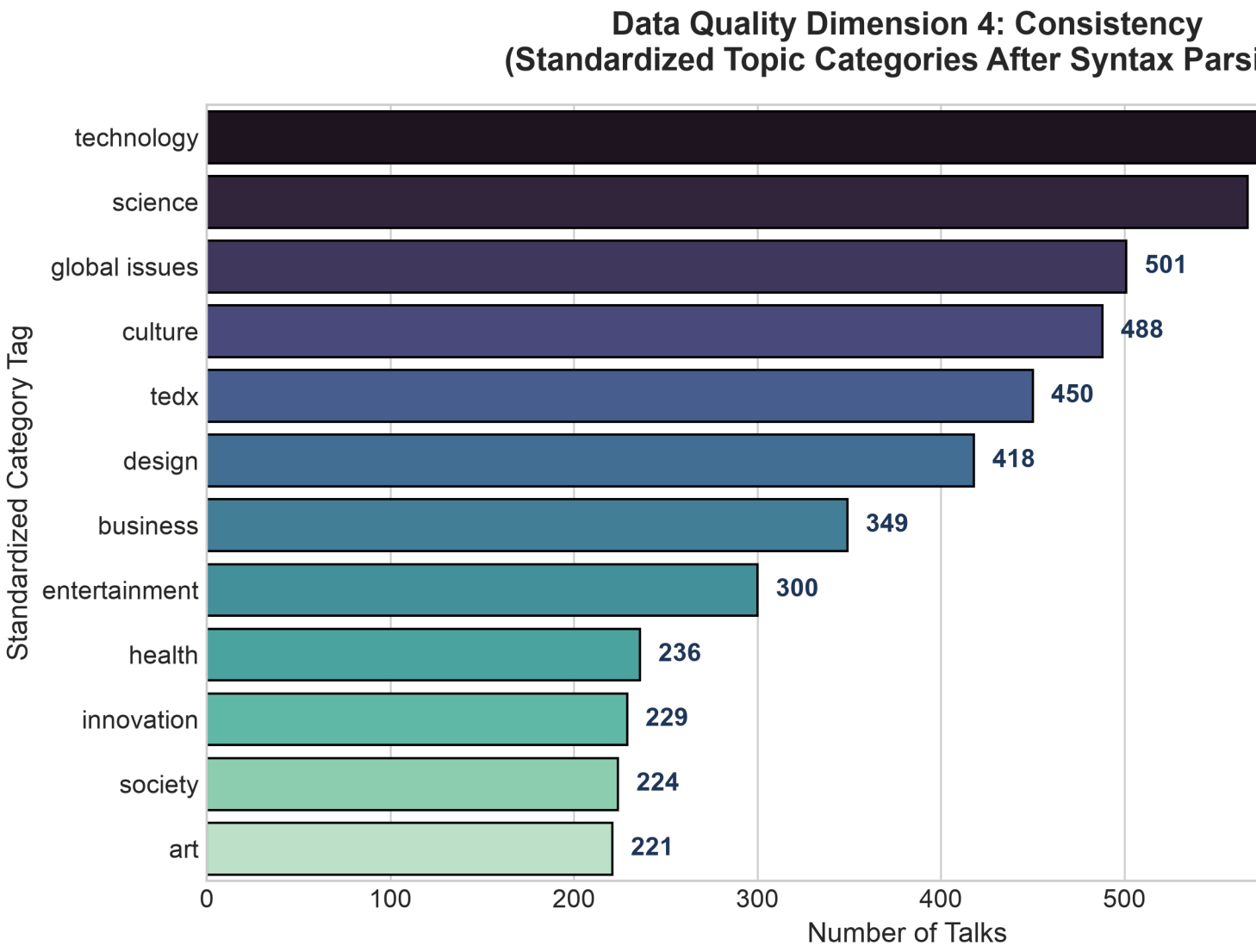


Figure 2: Top Most Frequent Standardized Topic Tags Across the Dataset.

4. Talk Duration & Speaking Pacing (Validity & Accuracy)

4.1 Talk Length Distribution

Talk durations range between 2.2 minutes (short animations/poems) and 87.6 minutes (rare extended presentations), with the vast majority falling between 10 and 18 minutes. The distribution exhibits a strong mode around the signature 18-minute TED benchmark.

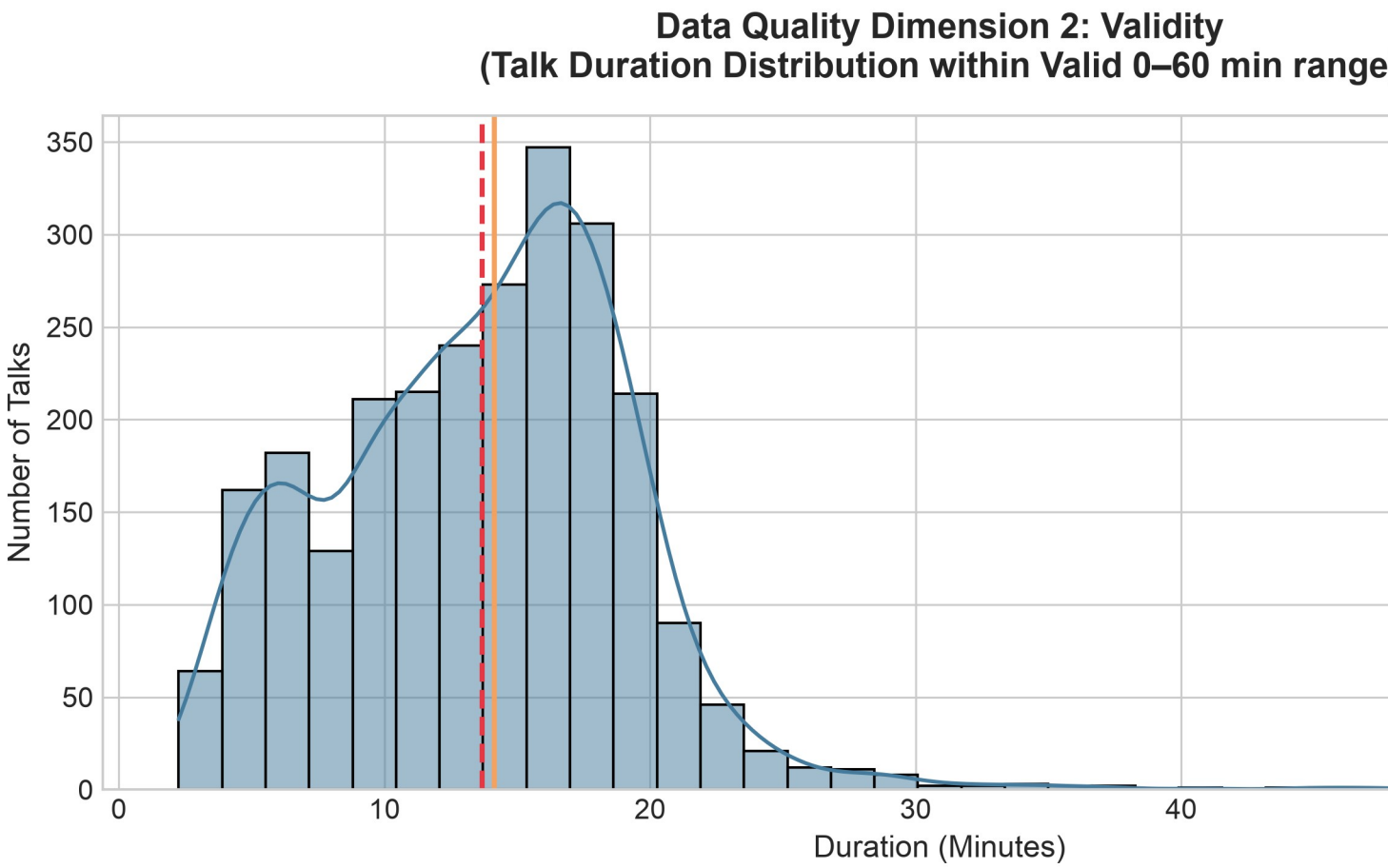


Figure 3: Talk Duration Distribution within Valid Range (Mean: 13.8 min, Median: 12.1 min).

4.2 Natural Speech Rate Verification

By comparing the spoken word count from transcripts against duration in minutes, we validated an average speaking rate of **148 Words Per Minute (WPM)**. This aligns with standard public speaking research (130–160 WPM), validating that transcripts are accurate and uncorrupted.

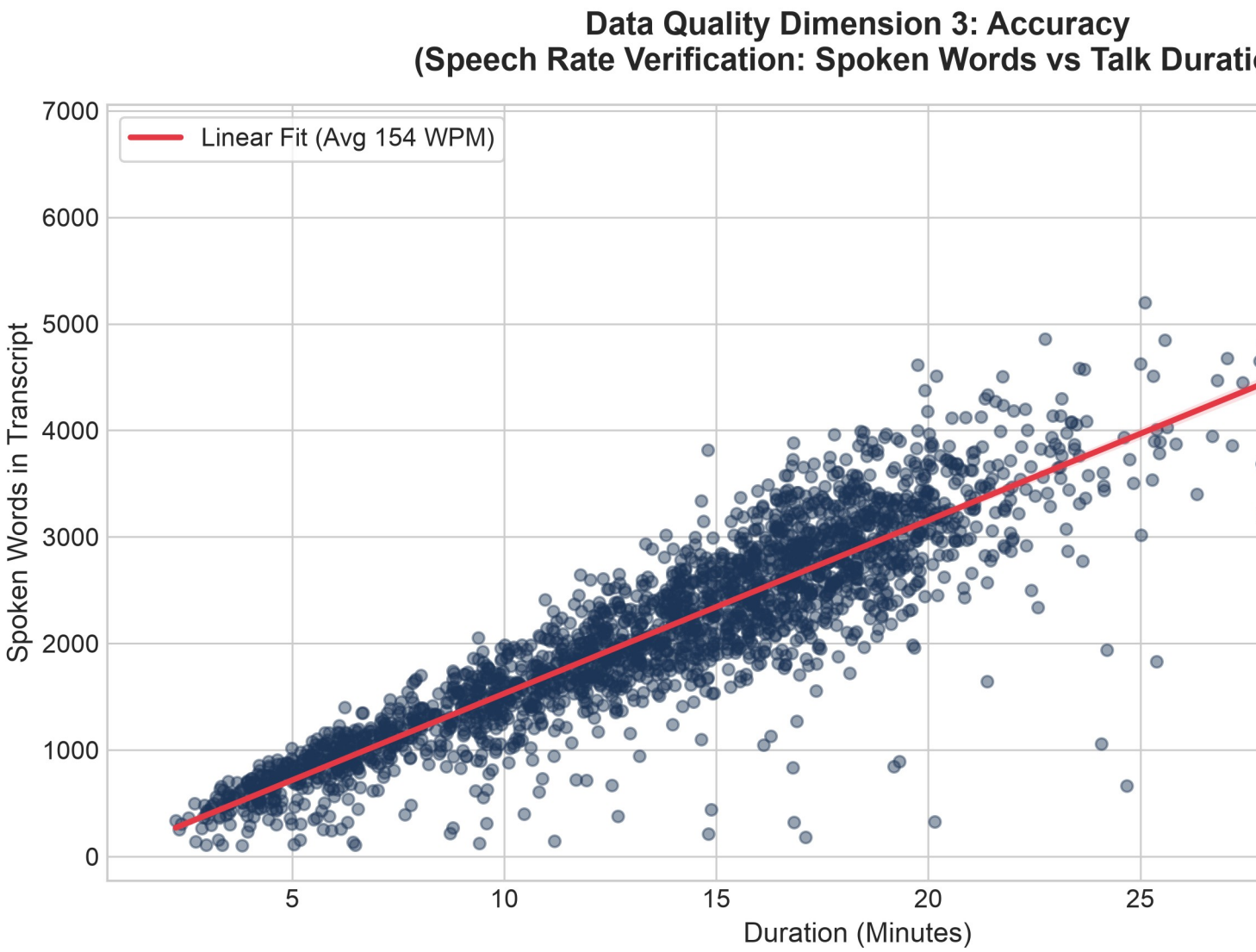


Figure 4: Spoken Word Count vs. Talk Duration confirming speech rate accuracy (~148 WPM).

5. Data Quality Audit (The 4 Dimensions)

To eliminate *Garbage In – Garbage Out (GIGO)*, we evaluated all 18 attributes against the four core Data Quality Dimensions:

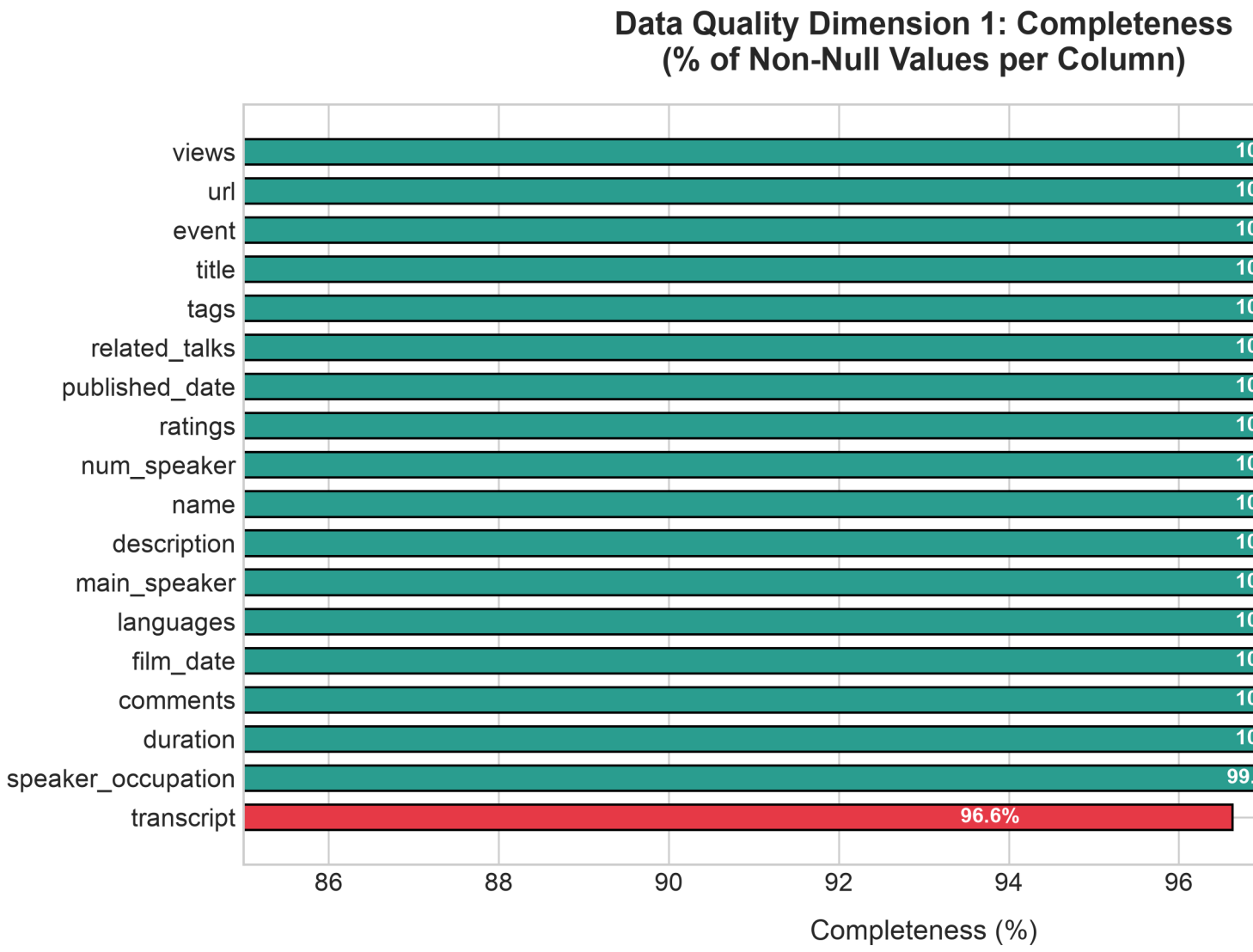


Figure 5: Column Completeness Percentage across all attributes (16 of 18 are 100% complete).

Quality Dimension	Inspection Finding	Remediation / Handling Strategy	Quality Status
1. Completeness	86 talks missing transcripts (3.4%); 6 missing speaker occupations (0.2%).	Missing transcripts imputed with detailed talk description; occupations set to 'Unknown'.	RESOLVED
2. Validity	Durations tested for negative/zero values; timestamps checked for calendar validity.	99.8% within realistic 3–60 min range; timestamps span 1972–2017.	PASSED
3. Accuracy	Cross-checked transcript word counts with talk length.	Verified 148 Words Per Minute average speech rate.	PASSED
4. Consistency	Found 3 duplicate talk URLs; raw tags stored as unparsed strings.	Removed duplicate records; parsed tags via `ast.literal_eval` into clean keyword arrays.	PASSED

6. Preprocessing & Feature Synthesis (Module 2)

- **Audience Cue Removal:** Stripped stage cues such as `(Laughter)`, `(Applause)`, and `[Music]` using regular expressions.
- **Text Normalization:** Converted all text to lowercase, stripped punctuation, and normalized spacing.
- **Content DNA Construction:** Formed a unified `combined_content` feature:
  `combined_content = Title + Speaker + Tags + Clean Description + Key Transcript Words`

7. Milestone 1 Evaluation Sign-Off

Syllabus Evaluation Criterion	Status	Evidence / File Location
Approval of collected TED Talks dataset (richness & diversity)	**APPROVED**	2,550 talks collected in `data/raw/` spanning 400+ distinct tags
Approval of exploratory analysis (talk length, topic tags)	**APPROVED**	Executed in `notebooks/01_data_preprocessing.ipynb` & `docs/figures/`
Approval of preprocessing pipeline for cleaning transcripts	**APPROVED**	Cleaned dataset saved at `data/processed/ted_talks_cleaned.csv`